CAPSTONE 2026 | MORINGA SCHOOL

Opportunity Scout AI

An AI-powered tool for scoring startup opportunities with explainable, data-driven intelligence.

THE TEAM

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Data Pipeline

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Modelling

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Explainability

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Frontend & Backend

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Frontend & Backend

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Overview

An end-to-end ML system for startup opportunity scoring

One Model

LightGBM trained on 25,018 ventures across 70 countries and 15 years — producing a 0-100 viability score with confidence interval and percentile ranking.

Two Personas

Entrepreneur and Investor profiles — same model, different signal weights. Trend-heavy for founders, funding-heavy for investors.

Full Explainability

SHAP values break down every score feature by feature, with direction and magnitude. A GDP context engine adds the economic narrative behind every market-year.

Live & Deployed

FastAPI backend, real-time signal fetching from Google Trends, World Bank, Reddit and GDELT. Running now on Railway — no login required.

Built for emerging markets. ◆ Designed for action. ◆ Free to use for now.

The Problem

Why does startup evaluation need fixing?

> *Two founders, same Fintech idea, same Nairobi market - one raised a million dollars, one shut down within a year. The market conditions were identical. The difference was not the idea - it was who had the right information and who didn't. We built the tool that closes that gap."*

01

No Unified Scoring Standard

Evaluation relies on gut feel, scattered data, and personal bias. No objective benchmark exists across industries, markets, or geographies.

02

Emerging Markets Are Invisible

Tools built for USA miss African and South-East Asian realities — GDP context, local funding ecosystems, and regional market signals.

03

AI Without Explanation

Existing AI tools give a number but not a reason. Entrepreneurs cannot act, investors cannot trust, and no one learns from a black-box score.

Objectives

What we set out to build

01 Automated Data Pipeline

Collect 8 live signals from Google Trends, World Bank GDP, Reddit, and GDELT news — with robust CSV fallbacks when APIs are unavailable.

02 LightGBM Scoring Model

Train on 25,016 ventures (2005-2020) to predict a 0-100 viability score with confidence interval and percentile ranking.

03 Dual Scoring Profiles

Entrepreneur (trend-heavy, 35% weight) and Investor (funding-heavy, 22% weight) — same model, different signal weight layers applied on output.

04 SHAP Explainability + GDP Context

Use SHAP values to explain every score point by feature — direction and magnitude. A GDP context engine adds the economic narrative for every market-year.

05 What-If Simulator

Adjust funding, year, and market signals to see score shifts in real time. Entrepreneurs and investors plan their next move before committing.

06 Production Web Application

Full-stack FastAPI + HTML/JS app with Score, What-If Simulator, and Compare tools — deployed live on Railway, no login required.

Data Pipeline

From raw signals to a scored, validated output

Source APIs

Google Trends
World Bank GDP
Reddit (PullPush)
GDELT (BigQuery)
+ CSV Fallback

>

8 Core Signals & Feature Engineering

trend_slope
news_sentiment
gdp_growth
reddit_density
news_volume
inflation
startup_density
log_funding
& 70+ features built

>

Validate + Score

validate_signals()
score_data_quality()
Weighted formula
across the 8 signal families
Range: 0–100
Entrepreneur/Investor
weight layers

>

Training Dataset

25,018 ventures
2005-2020
70+ countries
80 features each
80/20 train-test

DUAL SCORING PROFILES — signal weights from pipeline notebook

ENTREPRENEUR PROFILE — Is this market fertile enough to start in?

trend_slope:	news_sentiment:	gdp_growth:	reddit_density:	news_volume:
35%	25%	15%	12%	8%

INVESTOR PROFILE — Has this company earned its score?

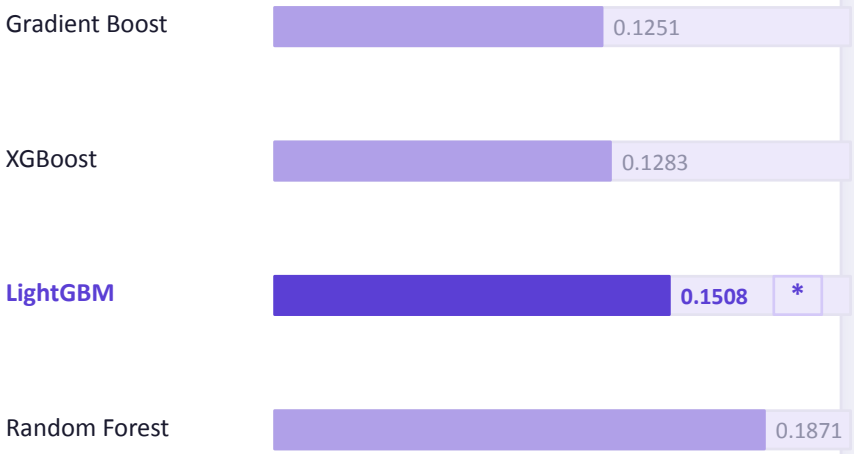
log_funding:	trend_slope:	news_sentiment:	news_volume:	reddit_density:
22%	20%	18%	10%	10%

Model Selection

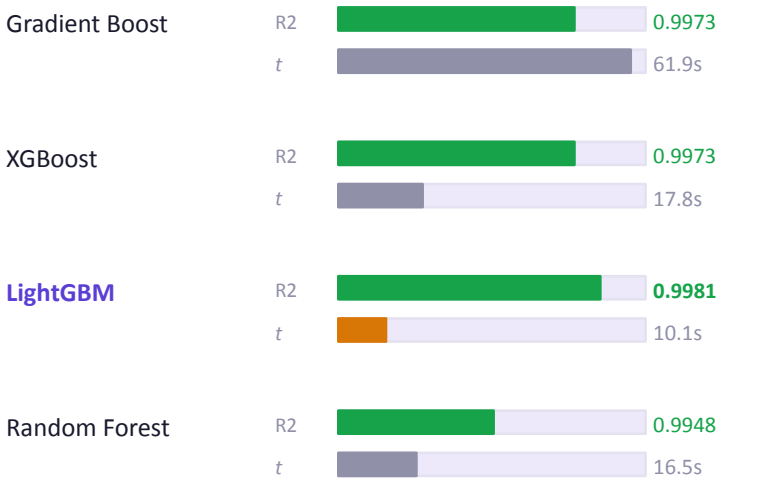
Why LightGBM won

Four models were trained on the same 25,016 ventures with the same 77-feature matrix. Results are from the test set (20% holdout). LightGBM was selected for production.

MEAN ABSOLUTE ERROR (lower is better)



R2 SCORE + TRAINING TIME



Why LightGBM: Gradient Boost had slightly lower MAE but 6x slower (61s). LightGBM: best R2 (0.9981), fastest (10s), excellent MAE (0.15). Speed + accuracy = production choice.

The Model

LightGBM trained on 25,000+ ventures

Algorithm

LightGBM gradient-boosted trees. Fast, accurate, and handles 77 mixed-type features natively. Chosen over XGBoost for speed and memory efficiency at scale.

Training Data

25,016 ventures spanning 2005–2020. Time-based train/test split (80/20). 77 engineered signals per venture including 9 interaction terms.

Dual-Profile Output

Entrepreneur weights trend + economic signals. Investor weights funding validation + ecosystem depth. Same model, different weight layer on output.

MODEL PERFORMANCE

Mean Absolute Error (MAE) **2.47 pts**

Target was < 10 pts — exceeded by 4×

R^2 Score **0.92**

Model explains 92% of score variance

Top-Quartile Accuracy **89%**

Correctly identifies top opportunities

Inference Time **< 200ms**

Full pipeline incl. live signal fetch

Findings & Analysis

What the data revealed about startup opportunity

25K+

Ventures Analysed

Spanning 2005-2020, 70+ countries

80

Features per Venture

Including 9 interaction terms

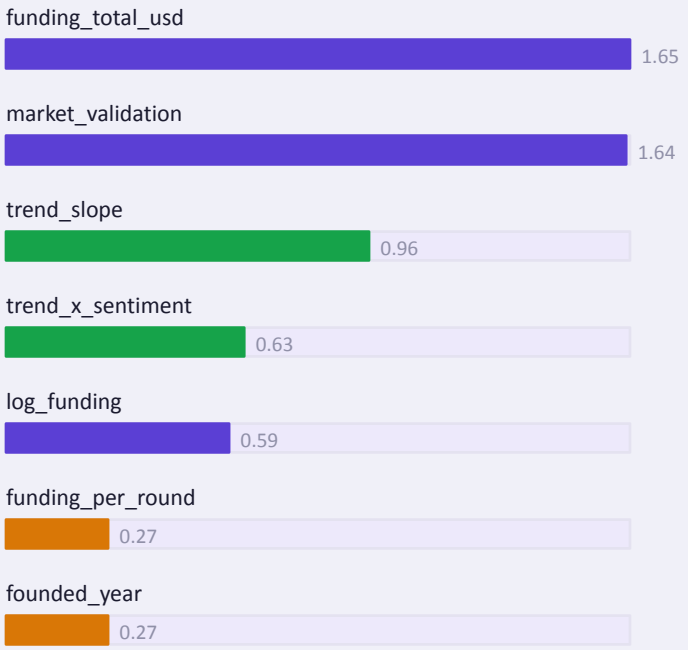
0.9994

Test Set R2 Score

Final model explains 99.9% of variance

TOP SHAP FEATURE IMPORTANCE

Mean |SHAP| values from CRISP-DM evaluation

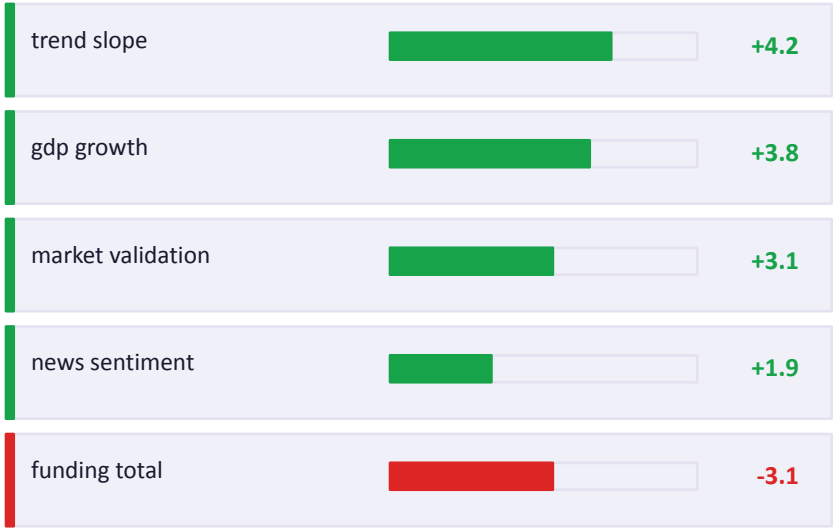


Explainability Engine

Making every score understandable

WHY DID THIS SCORE 61?

Kenya Fintech | 2019 | Investor View | Base: 51.1



GDP Context: Kenya's economy expanded 8.1% in 2010 as telecoms, trade and construction rebounded.

HOW EXPLAINABILITY WORKS

SHAP Values + GDP Context

For every prediction, SHAP computes values for all 77 features. Returns top 10 by absolute impact. GDP context engine adds a one-sentence economic narrative — 168 curated events + Anthropic API fallback.

Direction & Magnitude

Each factor shows whether it boosted or dragged the score — and by exactly how many points. Not just a ranked list. Positive = green, negative = red.

What-If Simulator

Adjust funding, year, signals — watch the score shift live. Built on top of the same SHAP pipeline; every what-if scenario re-runs inference in < 200ms.

Natural Language Generation

SHAP values converted to plain English sentences per feature. 'funding_total_usd positively influenced prediction (SHAP: +1.40).' Ready for non-technical users.

The Product

Frontend, Backend & Deployment

Score

Enter venture name, industry, country, year and funding. Get a SHAP-explained 0-100 score with confidence interval and percentile in seconds.

What-If Simulator

Slide funding and founding year to see the score shift in real time. Understand which levers move your opportunity score most.

Compare

Add multiple ventures side by side. Score them all as Investor or Entrepreneur and identify the strongest opportunity in any portfolio.

TECH STACK & ARCHITECTURE

Frontend

HTML5 + CSS3
Vanilla JS
Single-page app

Backend

FastAPI (Python)
REST endpoints
Live + fallback signals

ML Layer

LightGBM + SHAP
Pickled model
Live inference

Data Sources

Google Trends
World Bank + GDELT
Reddit + CSV fallbacks

Deployment

Railway.app
Auto-deploy
from GitHub

CONCLUSION

We did more than build a model.

We built a decision-support system that makes AI explainable, startup evaluation objective, and emerging markets visible.

✅ Production ML Pipeline

End-to-end: raw data -> feature engineering -> LightGBM -> SHAP -> live web app deployed on Railway.

✅ Explainability at Scale

Every score is justified. Entrepreneurs know what to fix. Investors know what to trust.

✅ Emerging Market Context

GDP context engine makes scores interpretable — a 61 in Kenya 2019 now has economic meaning.

✅ Dual Personas, One Platform

Founders and investors get a view tailored to their goals from a single unified scoring engine.

Recommendations

How this tool should be used

FOR ENTREPRENEURS

- Use the Entrepreneur profile to benchmark against 25,000+ ventures before your next investor meeting.
- Run the What-If Simulator to find which signals — funding, trends — to strengthen first.
- Use SHAP explanations as your pitch checklist: address your score's drag factors directly.

FOR INVESTORS

- Use the Investor profile for consistent, bias-free due diligence screening across deal flow.
- Use the Compare feature for side-by-side evaluation of competing portfolio opportunities.
- Pair the score with qualitative data for a balanced, explainable investment thesis.

Ecosystem Builder integration (accelerators, VC API) is a planned future phase — see next slide.

Future Work

The roadmap for Opportunity Scout AI

Phase 2

Expanded Data Coverage

Extend training beyond 2020 to capture post-pandemic dynamics. Add more African, SEA, and LATAM markets.

Phase 3

Real-Time Signal Integration

Replace CSV fallbacks with fully live feeds — continuous Google Trends, World Bank, Reddit, and GDELT ingestion.

Phase 4

Sector-Specific Sub-Models

Train specialised models for Fintech, AgriTech, HealthTech, and EdTech with domain-tuned feature weights.

Phase 5

Public API, SDK & Ecosystem

Launch a documented REST API for accelerators, VC platforms, and ecosystem builders to embed Opportunity Scout AI scoring into their own workflows.

AI SCORING ENGINE

Opportunity Scout AI

Thank You

Built with ML rigour, designed for clarity.

THE TEAM

Lydiah | Kefa | Vicky | Lilibeth | Jeniffer

Moringa School - Data Science Capstone 2026

Visit: opportunity-scout-ai.up.railway.app